

Human Capabilities

April 6, 2026 | week 2 | SWE 263P
Matthew Bietz

Today's Agenda

- ➔ What Sets HCD Process Apart? (~30 min)
- Human Capabilities & HCD + Break (~30 min + 15 min)
- Design Principles + Break (~30 min + 15 min)
- Activity (~30 min + 15 min)
- Coming Up (~5 min)

The Human-Centered Design Perspective

Blame computers, not humans! Computers are new,
humans are old

Our **human** instincts, bodies, capabilities are core to how
we interact with designed artifacts

Key **principles and practices** of (good and bad) design
undergird *all* design, including computing design

% of Software Projects that are...

On Budget



On Time



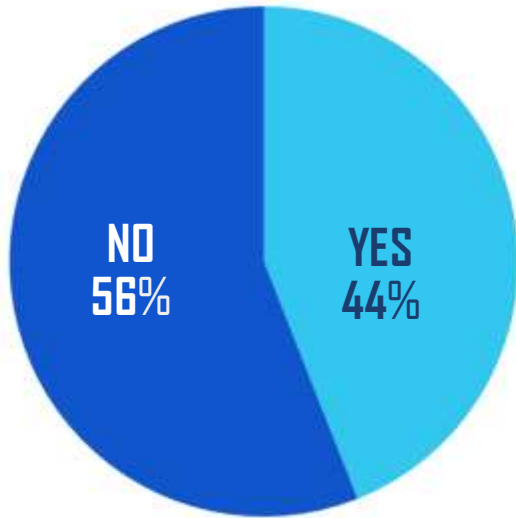
On Target



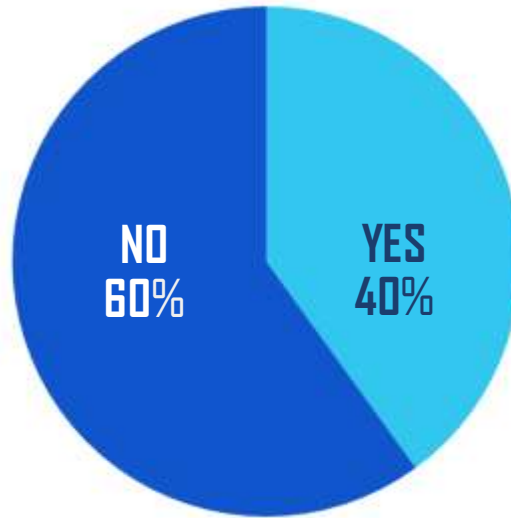
Standish Group Report, 2015

% of Software Projects that are...

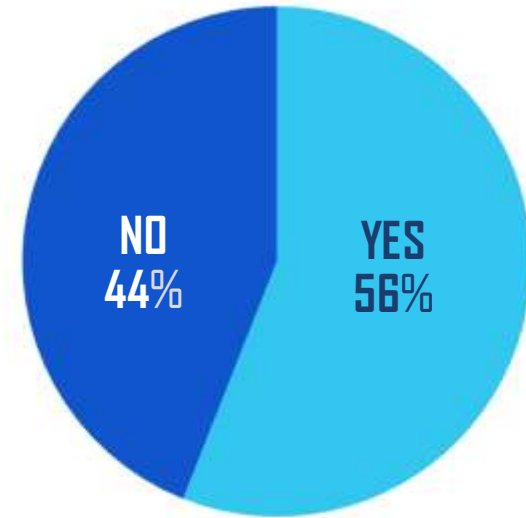
On Budget



On Time

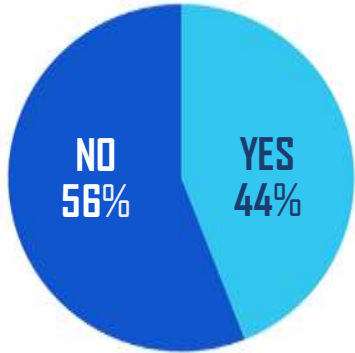


On Target

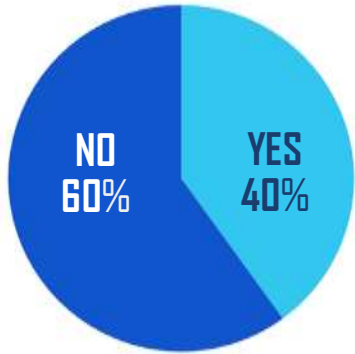


Standish Group Report, 2015

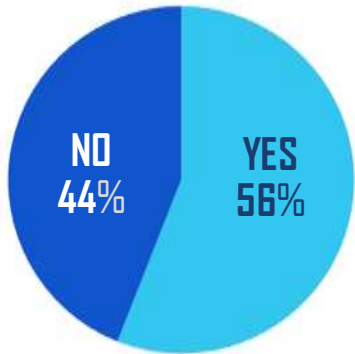
On Budget



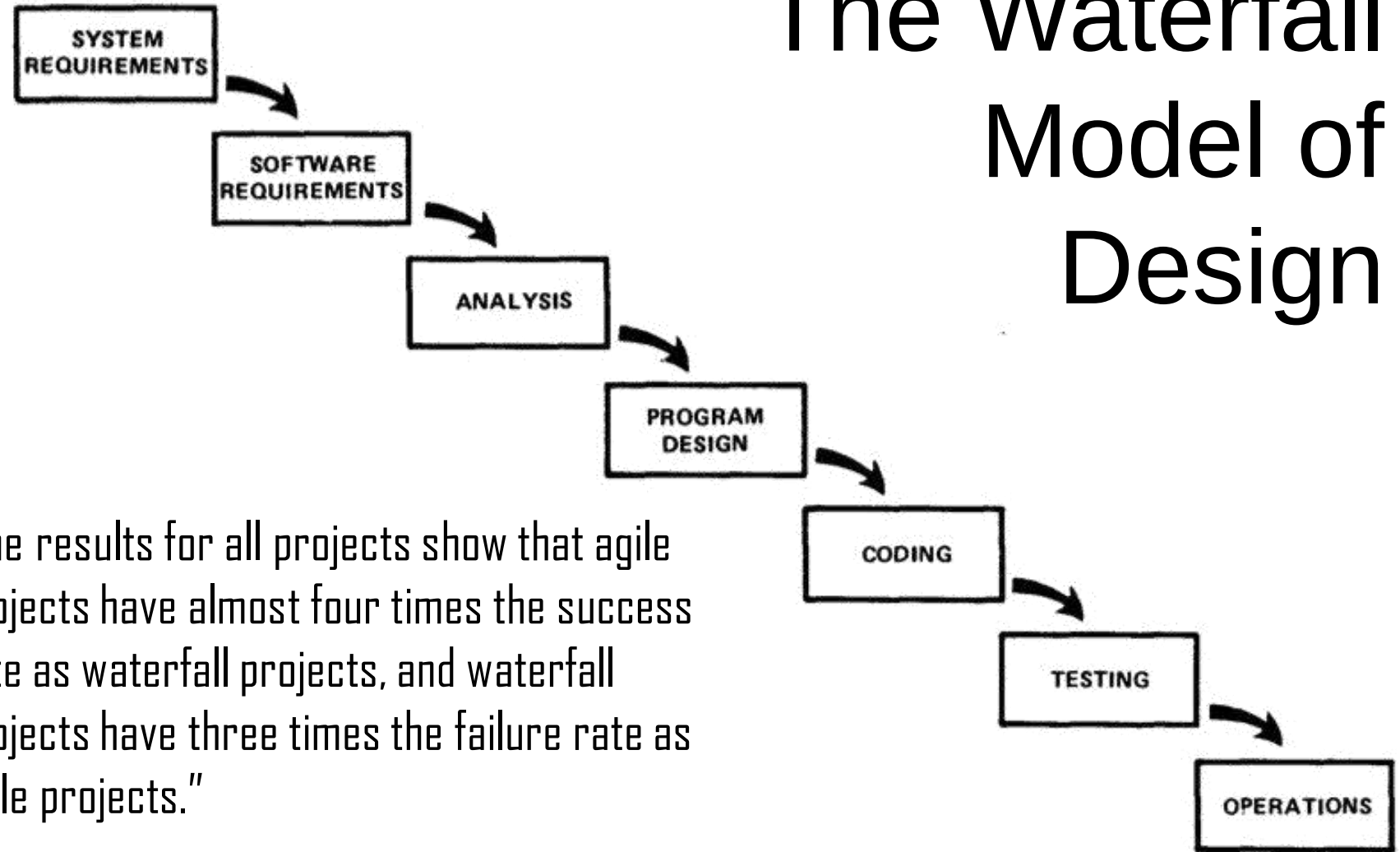
On Time



On Target



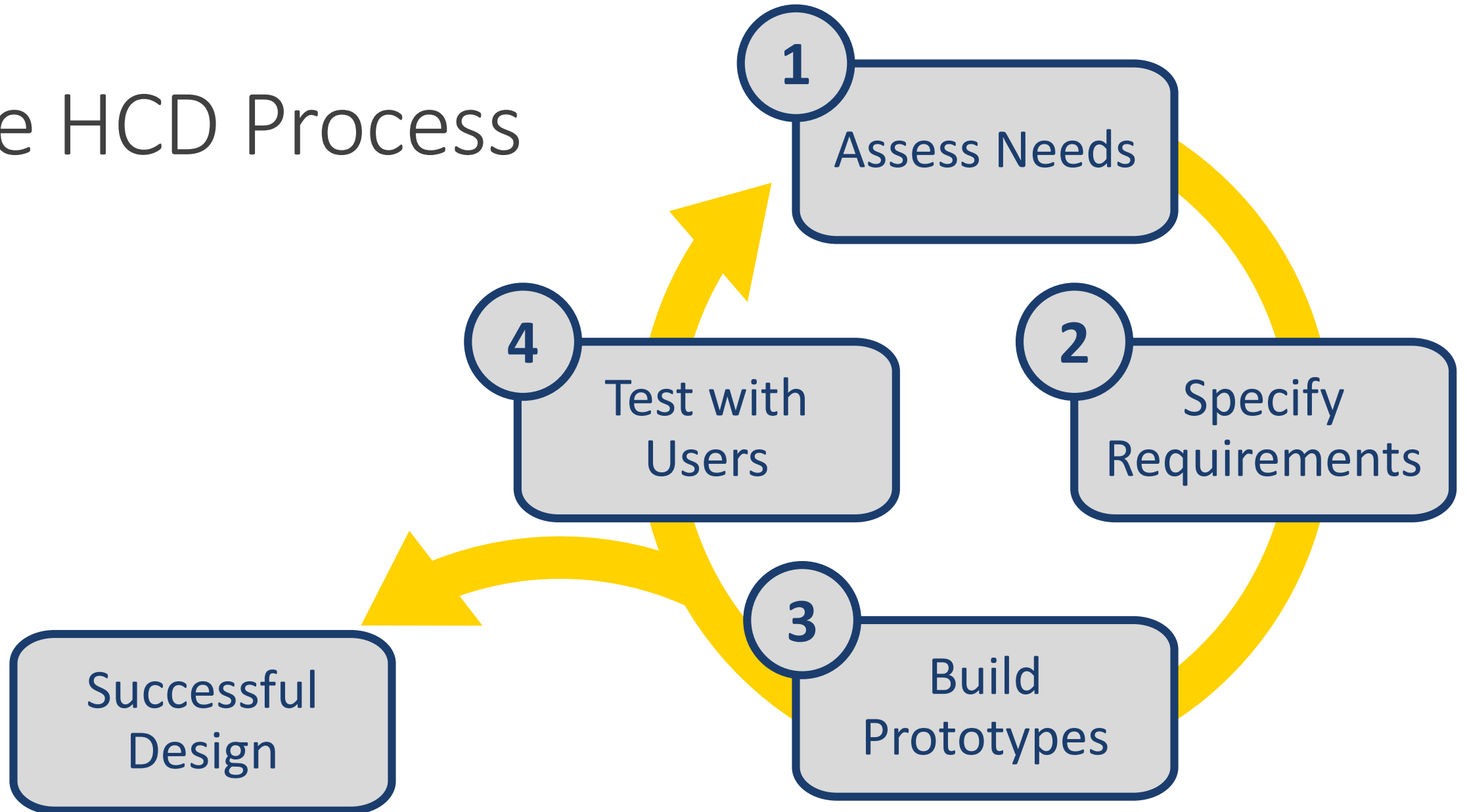
"The results for all projects show that agile projects have almost four times the success rate as waterfall projects, and waterfall projects have three times the failure rate as agile projects."



The Agile SE Process



The HCD Process



Matt's View on SE & UX Process Models

Models are usefult, but...

Abstraction is not reality

SE models tend to treat “design” as a black box

UX models tend to treat “implementation” as a black box

Tensions Between SE / UX Approaches

Temporalities

- Some UX processes don't fit 2-week sprints
- Some UX testing requires a working system
- Too-frequent releases (continuous delivery) can harm UX

Interdependence

- UX and software design affect each other

Different Subgoals

- Implement in increments vs. holistic user experience
- Software quality may mean different things
- Software often requires tradeoffs (example: 2-factor authentication)

IDEO's Design Process

[IDEO Design Process video on YouTube](#)

Qualities of HCD (in IDEO video)

Rapid prototyping – “fail often in order to succeed sooner”

Team environment – sharing, diverse perspectives

Spoke with / observed stakeholders – “real experts”

Props (baby doll, paper bags, plastic bags) + role play

Asked the stakeholders for feedback

Brainstorming Rules

No hierarchy

“Focused chaos”

Encourage wild ideas

As many ideas as possible

Defer judgment

Build on each others’ ideas

Today's Agenda

What Sets HCD Process Apart? (*~30 min*)

➔ Human Capabilities & HCD + Break (*~30 min + 15 min*)

Design Principles + Break (*~30 min + 15 min*)

Activity (*~30 min + 15 min*)

Coming Up (*~5 min*)

Designing for Human Capabilities

Basic Principle:

Designs that are matched to human cognitive and physical capabilities will be easier to use and provide a better user experience.

Memory Experiment

Close your computers & put down your phones
(just for a couple minutes)

Focus on the presentation screens

R T J V P

Recite The Letters

W X Y I M C P

T X O P I Y

Q A H E W M

Recite the Letters

Cow Dog

Bird Cat

Rabbit

List the Animals

WHICH SET WAS EASIEST TO REMEMBER?

Short-Term vs. Long-Term Memory

Can't remember things for long without practice

Can keep a limited number in short-term memory: “7 plus or minus 2” (Miller’s “magic number”)

Grouping and hierarchy

Mnemonics

- “My Very Educated Mother Just Served Us Nachos”



Photo by [Flood G.](#)



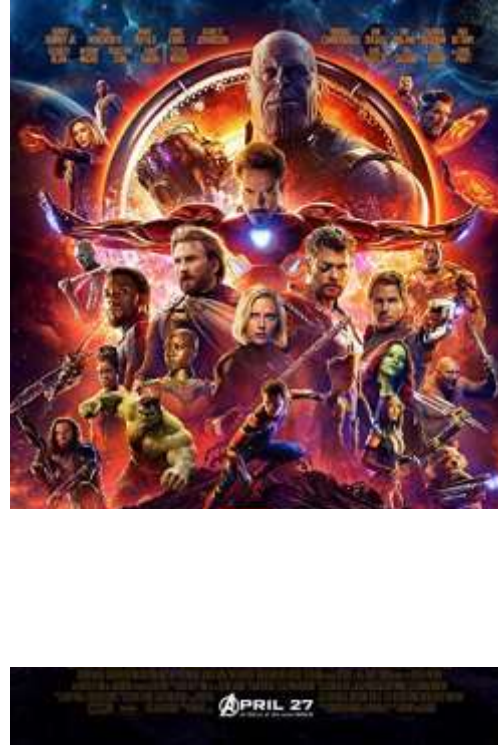
What 2022 film won the Best Picture Oscar at the 2023 Academy Awards?

A: Everything Everywhere All at Once

Which is a poster for a Marvel Movie?



A



B



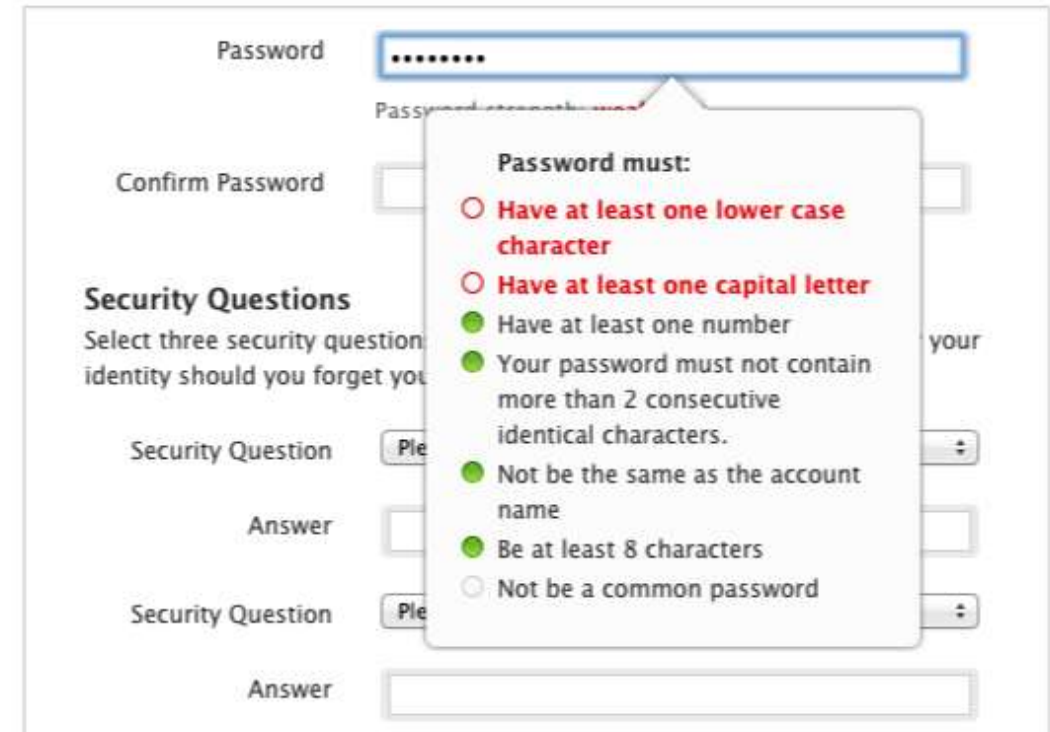
C

Recognition vs. Recall

Human memory and computer memory work in different ways

- Humans are better at **recognition**
- Computers are better at **recall**

We can use this kind of knowledge to design more intuitive systems



The image shows a portion of a web form for creating a password. It includes fields for 'Password' (masked with dots), 'Confirm Password', and 'Security Questions' (with dropdown menus for questions and text boxes for answers). A tooltip is displayed over the 'Password' field, titled 'Password must:', listing the following requirements:

- ☐ Have at least one lower case character
- ☐ Have at least one capital letter
- ☒ Have at least one number
- ☒ Your password must not contain more than 2 consecutive identical characters.
- ☒ Not be the same as the account name
- ☒ Be at least 8 characters
- ☐ Not be a common password



The image shows the interface of the '1Password Strong Password Generator'. The title is 'Need a password? Try the 1Password Strong Password Generator.' Below the title, it says 'Generate secure, random passwords to stay safe online.' The generated password is displayed in a white box: 'pascal - legacy - bumpy - mignon'. Below the password box, there is a dropdown menu set to 'Memorable Password' and a green button labeled 'Copy Secure Password'. At the bottom, there is a 'Length' slider set to 4, and checkboxes for 'Capitalize' (unchecked) and 'Full Words' (checked).

15 min break

Today's Agenda

What Sets HCD Process Apart? (*~30 min*)

Human Capabilities & HCD + Break (*~30 min + 15 min*)

 Design Principles + Break (*~30 min + 15 min*)

Activity (*~30 min + 15 min*)

Coming Up (*~5 min*)

Norman Doors

[Norman Doors video on
YouTube](#)

Norman's 6 Design Principles

Affordances

Signifiers

Constraints

Mappings

Feedback

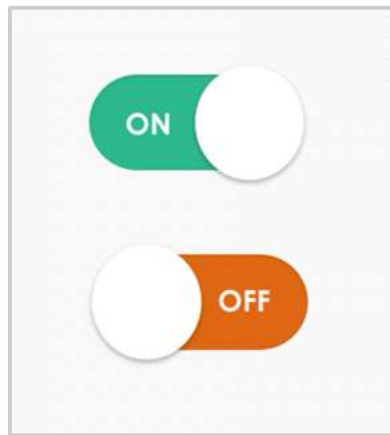
Conceptual Model



Photo by [Alfred Kenneally](#)

Affordances

The user should be able to discern **what actions they can take** on the interface – what it *affords*.



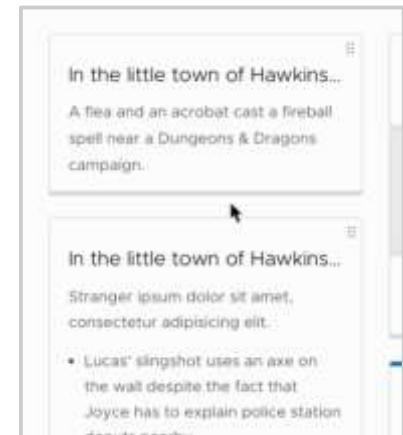
a switch affords
flipping



a wheel affords
turning



a button affords
pressing



a grip affords
dragging

Signifiers

Good signifiers **communicate** what the affordances of the interface are

To visit the link, click here ←

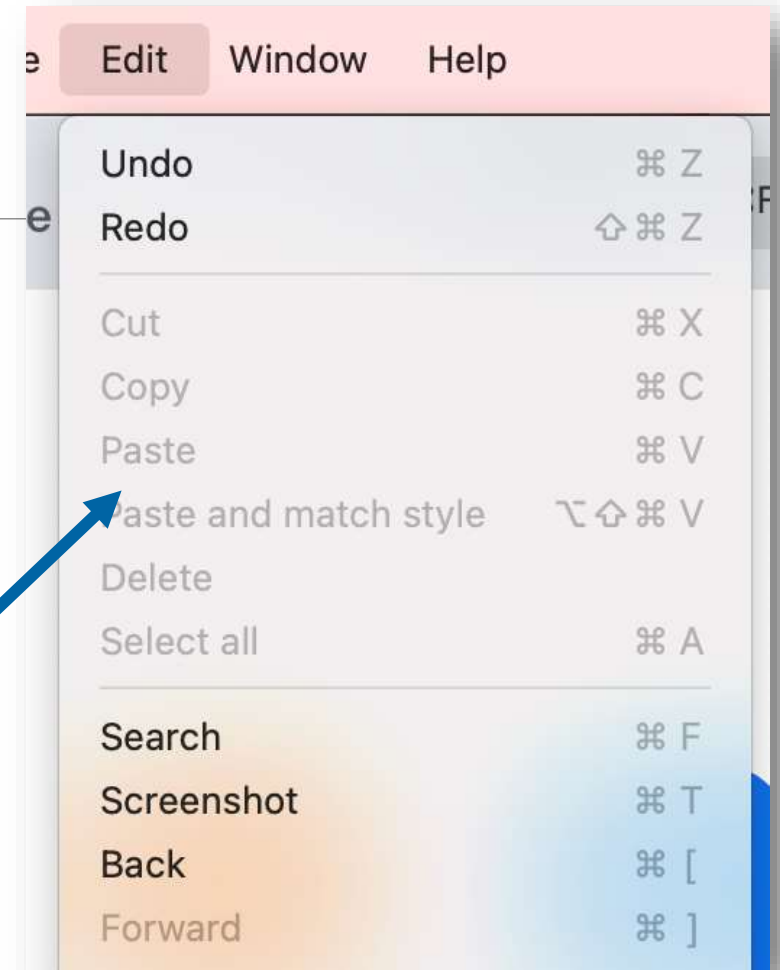
Affordance: I can click it

Signifier: Blue text with underline

Constraints

Limiting the user's **possible actions** with the system to reduce complexity and likelihood of error; constraints can be physical or digital.

Menu items that aren't currently available are grayed out and can't be clicked.



Mapping

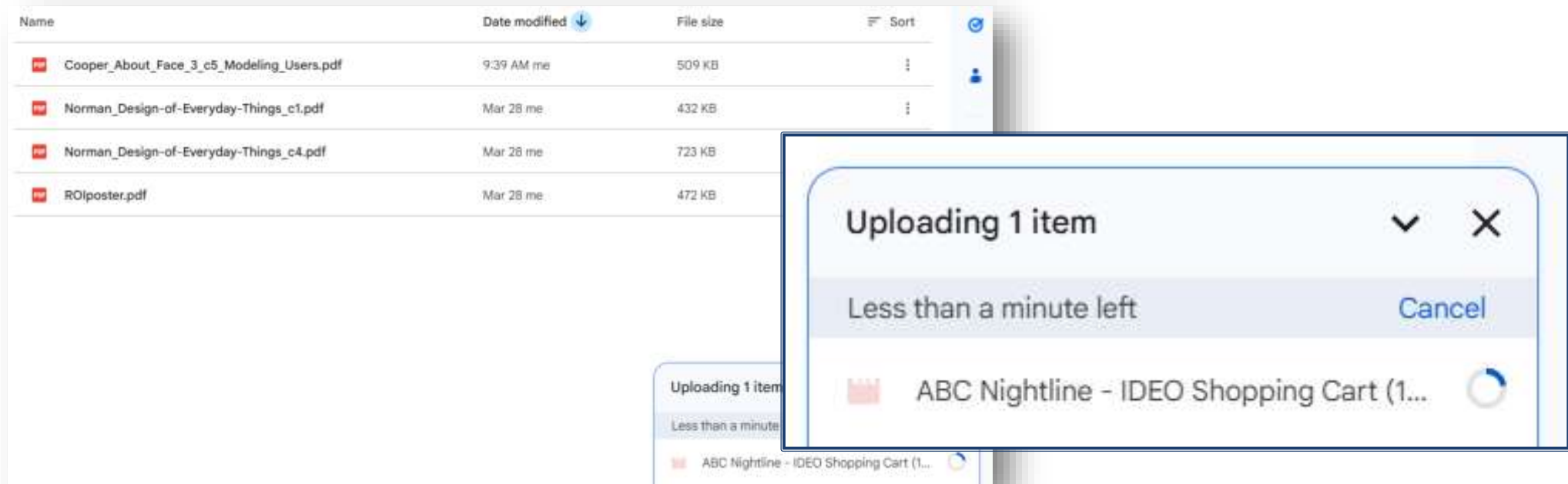
It should be easy to **predict what reaction** the system will have when the user interacts with a system control.

[click here](#) X

visit [Canvas](#) ✓

Feedback

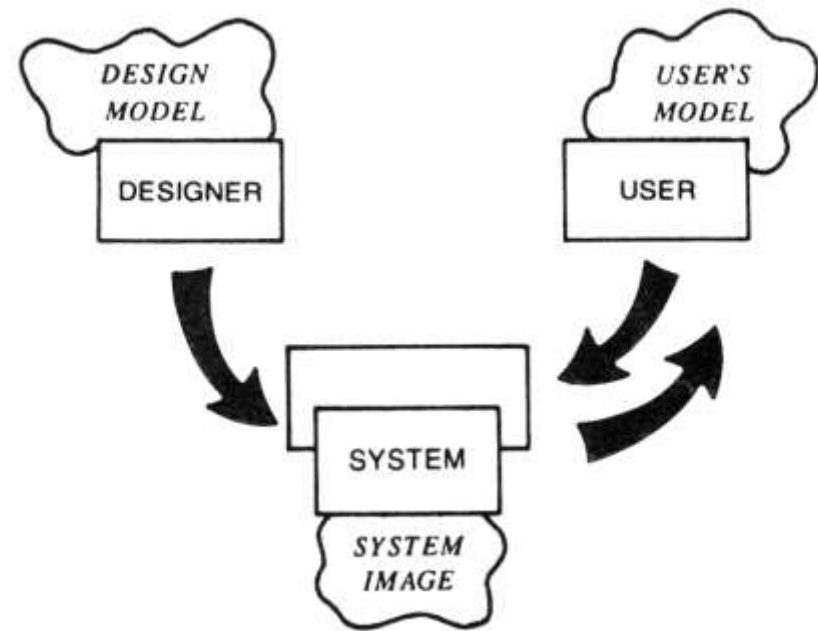
Provide information back to users that indicates their **action was registered** by the system; feedback should be given for both actions and status.



Conceptual Model

Designers have to communicate their **conceptual model** to the user through the interface (system image)

The user refines their **mental model** through interacting with the system over time



DESIGNER
CONCEPTUAL
MODEL



E-BOOK
READER



DESIGNER

USER
MENTAL
MODEL



TRADITIONAL
BOOK



USER

USER
MENTAL
MODEL



TRADITIONAL
BOOK

DESIGNER
CONCEPTUAL
MODEL



E-BOOK
READER

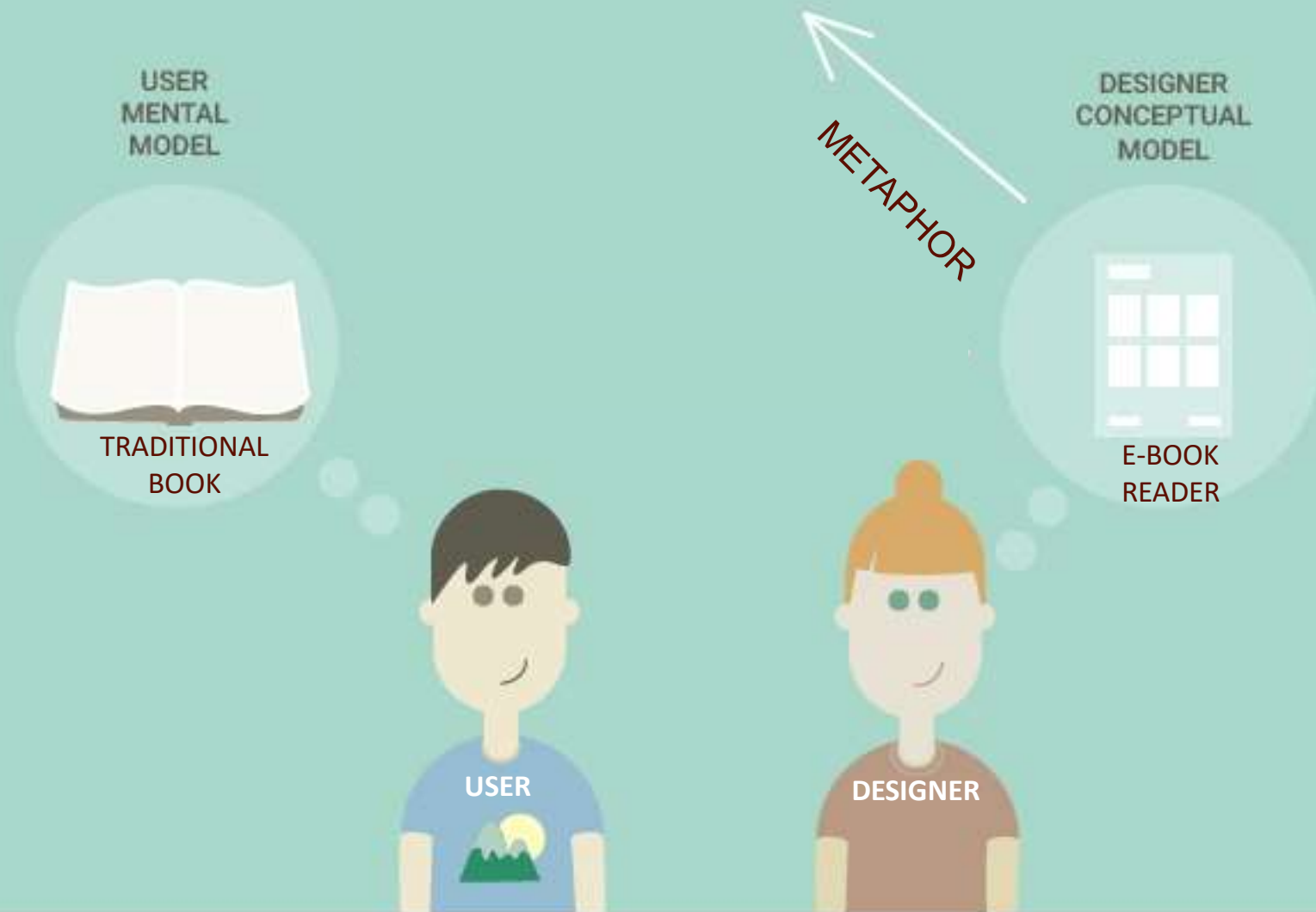


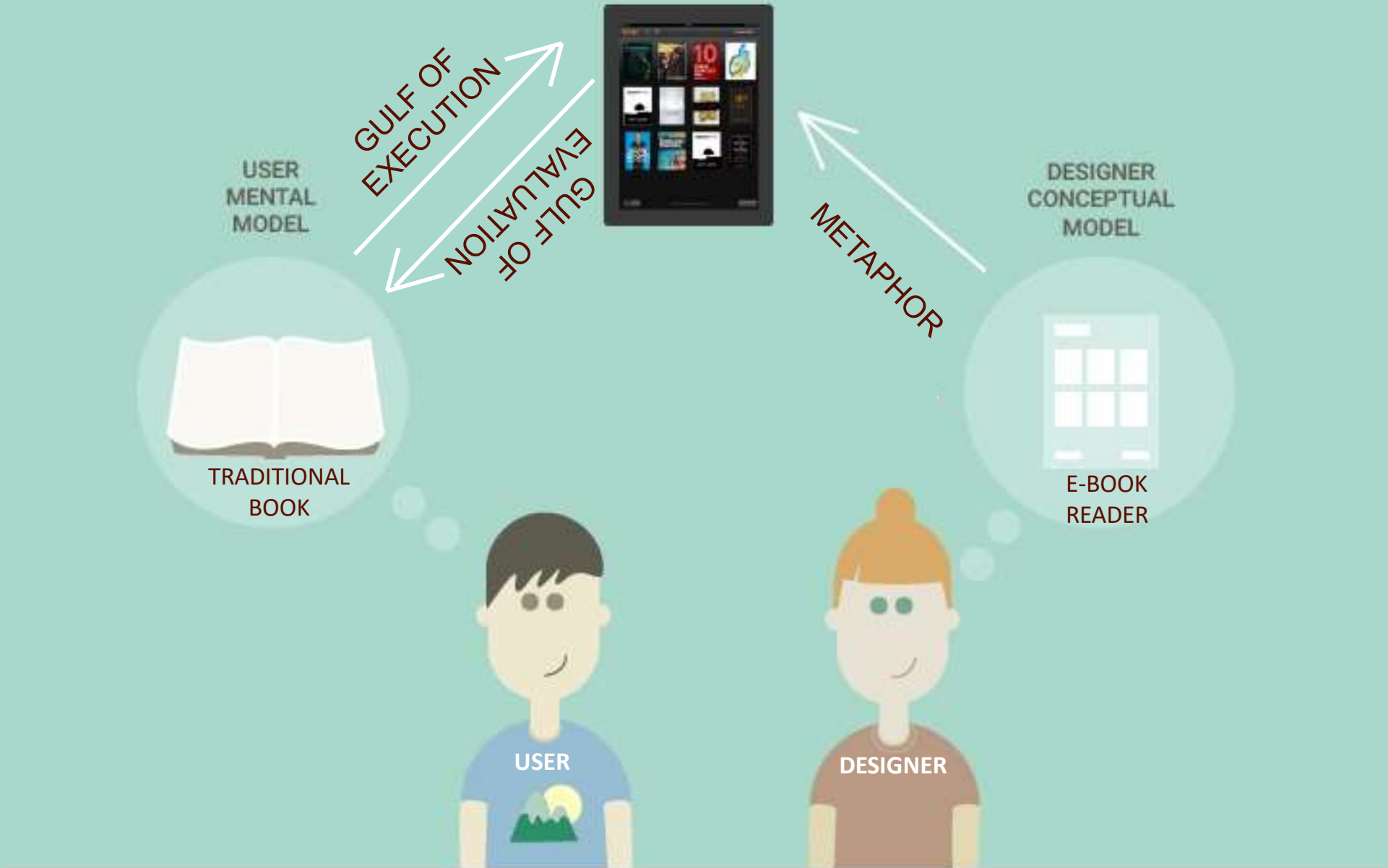
USER



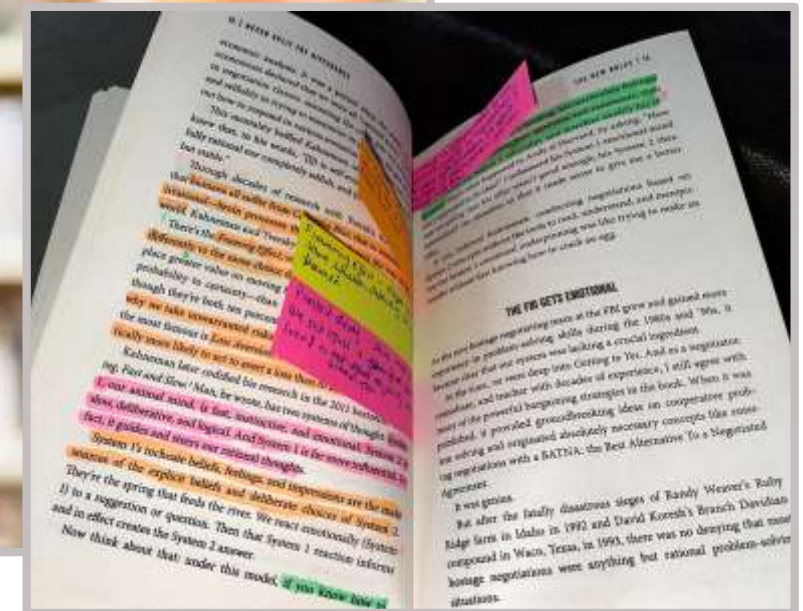
DESIGNER



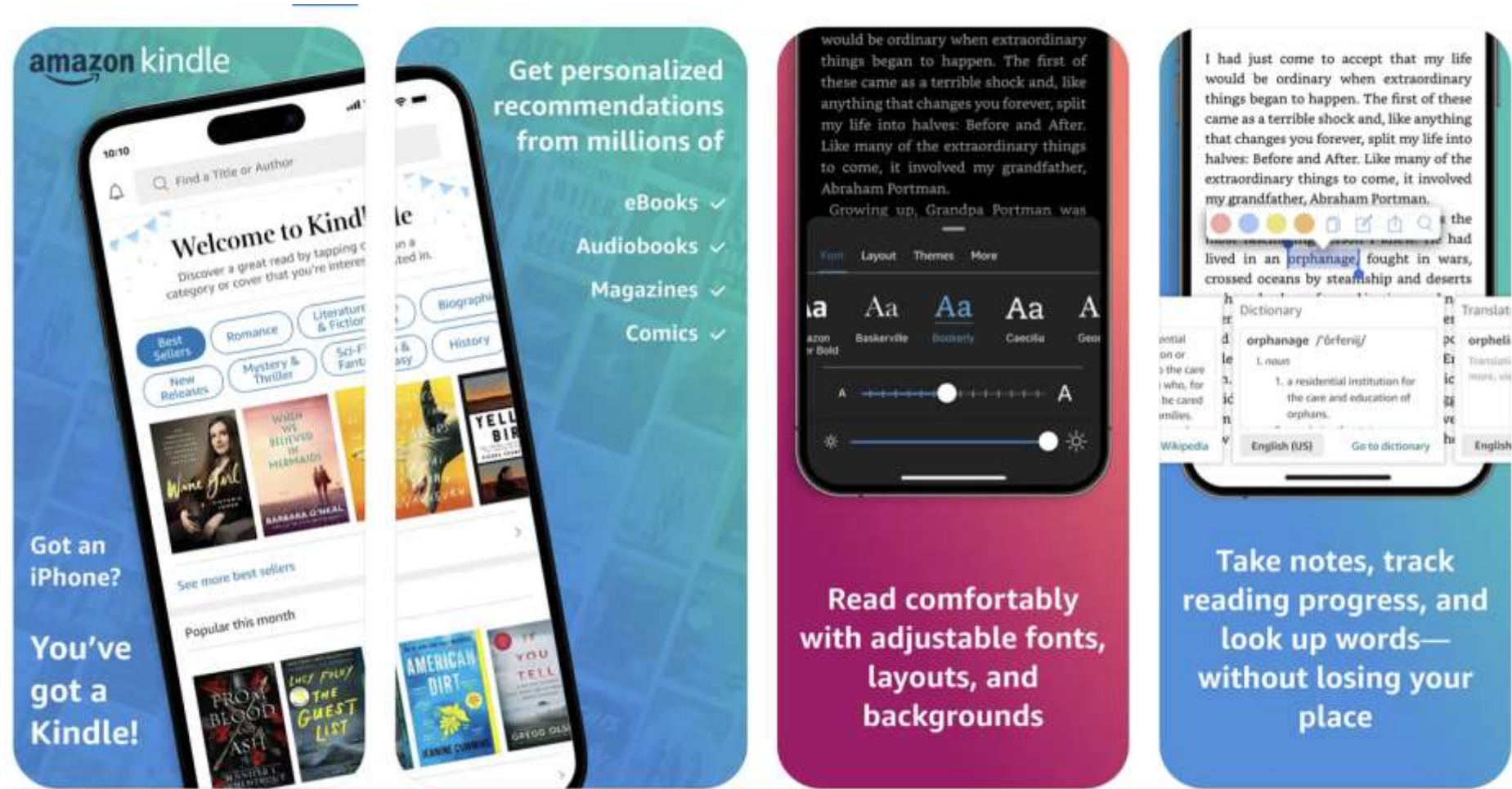




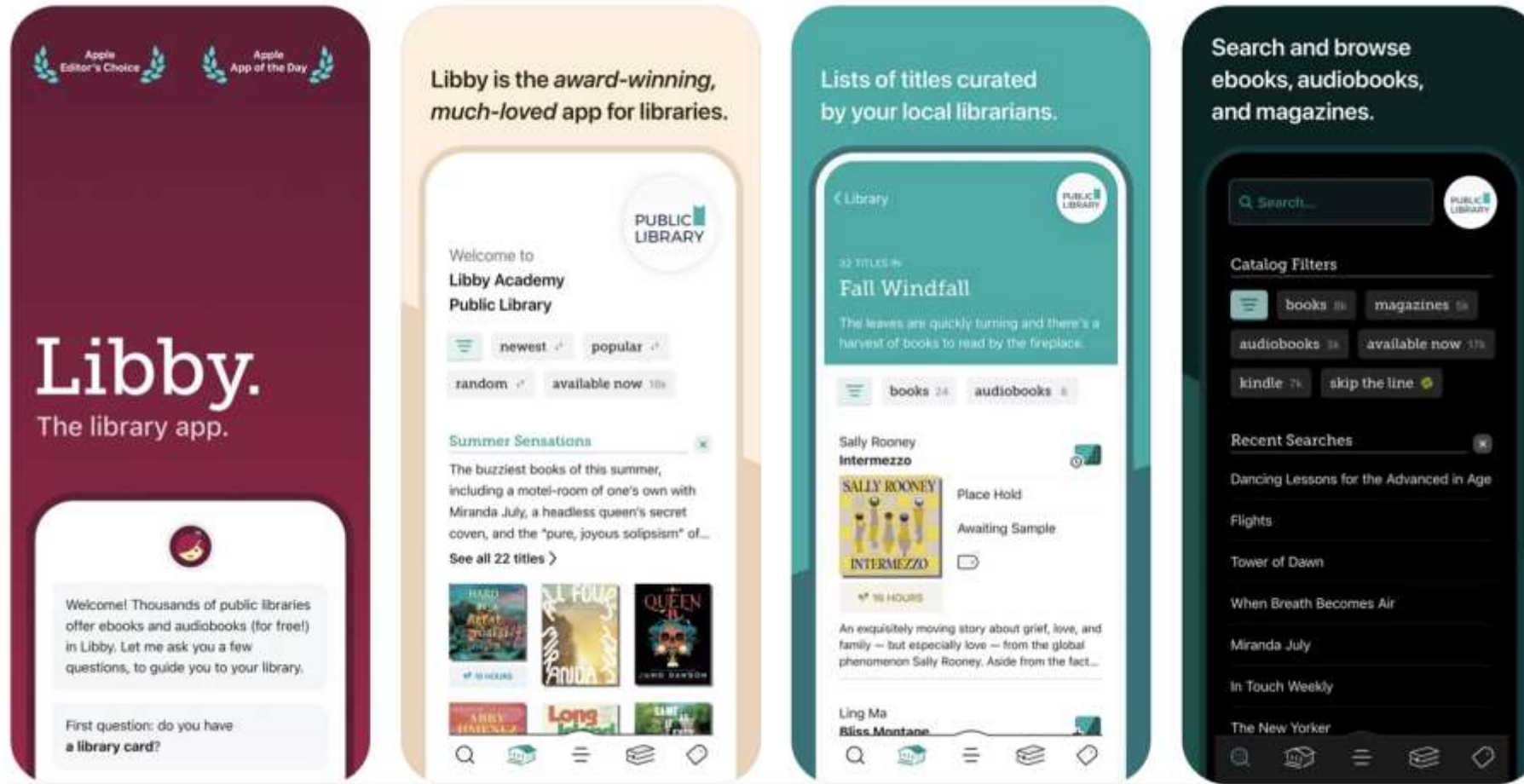
Mental Model: Books / Libraries



Conceptual Models: Personal Library



Conceptual Models: Public Library



Challenge of Designing for Mental Models

Mental models are adopted from **prior experience** with systems and technologies, e.g., “The same stimuli were grouped together in version x, so...”

Mental models are **age-based, culturally situated**, etc., so you have to design to multiple mental models to support more users

Your challenge as designer: determine the user’s Mental Model (but remember – **“The User Is Not Like Me”**)

Norman's 6 Design Principles

Affordances: Suggest what action can be taken

Signifiers: Indicate where the actions can take place

Constraints: Limit possible actions to prevent error

Mappings: Suggest what the result of actions will be

Feedback: Indicate system status and response to input

Conceptual Model: Representation of how the system works

10 min break

Today's Agenda

What Sets HCD Process Apart? (~30 min)

Human Capabilities & HCD + Break (~30 min + 15 min)

Design Principles + Break (~30 min + 15 min)

 Activity (~30 min + 15 min)

Coming Up (~5 min)

In-Class Activity: Fundamental Principles

Form a group of 3-4 people

Analyze how a Google form does or does not support Norman's design principles

Add a slide to share your findings

Links are available on the Canvas Week 2 page.

Assignment: “A Thing I Hate”

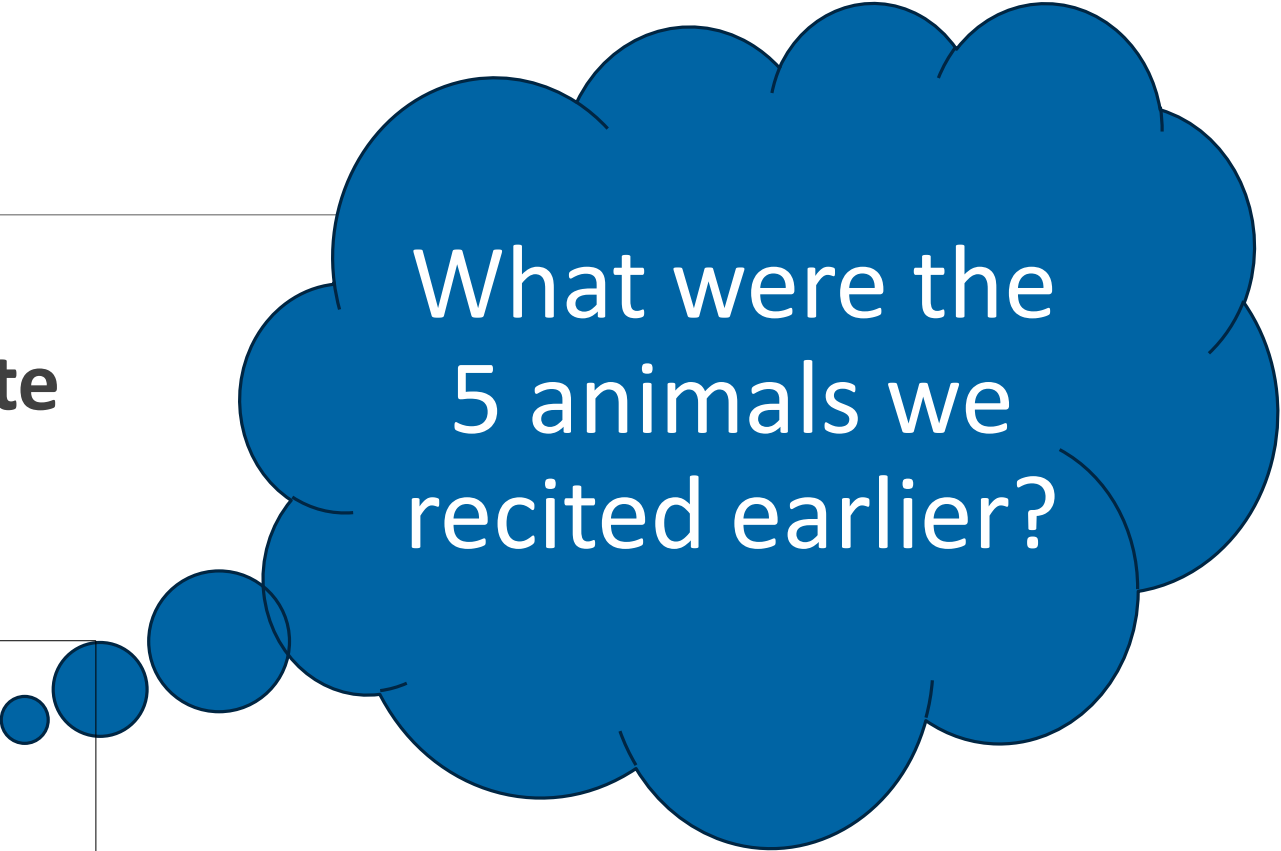
Identify and analyze some designed object (a physical object, a computer interface, etc.) that you personally find frustrating.

Use the ideas that we read about in Norman.

Write 400-500 words and submit by Sunday, 11:59pm.

This week

- Complete Week 2
Deliverable **A Thing I Hate**
- Do Week 3 **readings**



What were the
5 animals we
recited earlier?

The picture can't be displayed.